

Notes: Hall & Horowitz (Econometrica, 1996)

Bootstrap Critical Values for Tests Based on Generalized-Method-of-Moments Estimators

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1 Big picture

- **Question.** Can the bootstrap improve finite-sample critical values for tests based on generalized-method-of-moments (GMM) estimators, especially when data are dependent?
- **Motivation.** First-order asymptotic critical values for GMM tests can be very inaccurate in finite samples. Monte Carlo studies before this paper found large gaps between nominal and true rejection probabilities for both t tests and the overidentifying-restrictions test.
- **Main theoretical contribution.** The paper gives conditions under which bootstrap critical values provide asymptotic refinements for GMM-based t tests and the GMM J test. In their notation, bootstrap tests have level error $o(n^{-1})$ rather than the larger errors associated with first-order approximations.
- **Main practical complication.** With dependent data, one cannot simply resample individual observations. The bootstrap must sample blocks, and the bootstrap versions of the t and J statistics are not obtained by naively applying the sample formulas to the bootstrap sample.
- **Why naive bootstrap statistics fail.** Under block resampling, the conditional covariance of bootstrap sample moments differs from the covariance estimator that appears in the usual GMM statistics. Therefore the naive bootstrap t statistic does not have conditional variance one, and the naive bootstrap J statistic is not conditionally chi-square to the required order.
- **Corrected bootstrap statistics.** Hall and Horowitz derive corrected bootstrap versions of the t and J statistics. The correction rescales the bootstrap statistic using the covariance induced by the block bootstrap, not simply the covariance formula used in the original statistic.
- **Dependent data.** The paper focuses on stationary, ergodic time series and imposes strong mixing, smoothness, tail, and Cramer-type conditions that allow Edgeworth expansions with dependent observations.
- **Monte Carlo evidence.** In a nonlinear moment-condition model, bootstrap critical values usually reduce finite-sample level distortions, especially for symmetric two-sided t tests. Bootstrap critical values also indicate when first-order asymptotic critical values are unreliable.
- **Takeaway.** The bootstrap can deliver higher-order improvements for GMM inference, but with dependent data it must be implemented carefully. Correct recentering, block resampling, and covariance corrections are essential.

2 Model, data structure, and GMM objects

2.1 Moment condition and parameter

The GMM problem is based on the population moment condition

$$\mathbb{E}[g(X, \theta_0)] = 0, \quad (1)$$

where

- $X \in \mathbb{R}^{L_x}$ is the observed random vector;
- $g(X, \theta)$ is an $L_g \times 1$ vector of moment functions;
- $\theta \in \Theta \subset \mathbb{R}^{L_\theta}$ is the parameter;
- $L_g \geq L_\theta$, so the model can be just-identified or overidentified;
- θ_0 is the true parameter value.

Interpretation: This is the standard GMM setup. The paper is not about a new estimator; it is about the accuracy of tests built from GMM estimators.

2.2 Dependence and construction of the sample

The observed data are denoted

$$\{X_i : i = 1, \dots, n\}. \quad (2)$$

The process is stationary and ergodic. The paper assumes that the covariance of moment functions vanishes beyond a finite lag K :

$$\mathbb{E}[g(X_i, \theta_0)g(X_j, \theta_0)'] = 0 \quad \text{if } |i - j| > K. \quad (3)$$

To write long-run covariance estimators as sample moments, the authors define

$$N = n - K, \quad \mathcal{X}_i = (X_i, \dots, X_{i+K})'. \quad (4)$$

Interpretation: This notational device lets the covariance estimator be treated as a smooth function of sample averages. That is important because Edgeworth expansions are applied to functions of sample moments.

2.3 One-step GMM estimator

The first estimator uses a fixed positive semidefinite symmetric weight matrix Ω and solves

$$\hat{\theta}_N = \arg \min_{\theta \in \Theta} \left[N^{-1} \sum_{i=1}^N g(X_i, \theta) \right]' \Omega \left[N^{-1} \sum_{i=1}^N g(X_i, \theta) \right]. \quad (5)$$

Interpretation: This is the simple GMM criterion with a nonrandom weight matrix. The fixed-weight case is easier to analyze, but the paper also covers the two-step estimator.

2.4 Two-step GMM estimator

The two-step estimator uses an estimated optimal weight matrix. First compute the one-step estimator. Then define a covariance estimator

$$\hat{\Omega}_N(\theta) = N^{-1} \sum_{i=1}^N \left[g(X_i, \theta)g(X_i, \theta)' + \sum_{j=1}^K H(X_i, X_{i+j}, \theta) \right], \quad (6)$$

where

$$H(X_i, X_{i+j}, \theta) = g(X_i, \theta)g(X_{i+j}, \theta)' + g(X_{i+j}, \theta)g(X_i, \theta)'. \quad (7)$$

The two-step estimator solves

$$\hat{\theta}_N = \arg \min_{\theta \in \Theta} \left[N^{-1} \sum_{i=1}^N g(X_i, \theta) \right]' \hat{\Omega}_N(\bar{\theta}_N)^{-1} \left[N^{-1} \sum_{i=1}^N g(X_i, \theta) \right], \quad (8)$$

where $\bar{\theta}_N$ is the first-step estimator.

Interpretation: The two-step estimator uses an estimated long-run covariance matrix. Its asymptotic distribution is standard, but higher-order analysis is harder because the weight matrix is random.

2.5 Derivative matrix and asymptotic covariance

Let

$$D = \mathbb{E} \left[\frac{\partial g(X, \theta_0)}{\partial \theta'} \right], \quad D_N = N^{-1} \sum_{i=1}^N \frac{\partial g(X_i, \hat{\theta}_N)}{\partial \theta'}. \quad (9)$$

For the fixed-weight estimator, the asymptotic covariance is

$$\Sigma = (D' \Omega D)^{-1} D' \Omega \Omega_0 \Omega D (D' \Omega D)^{-1}, \quad (10)$$

where Ω_0 denotes the long-run covariance of the moment vector.

For the two-step efficient GMM estimator, the asymptotic covariance is

$$\Sigma = (D' \Omega_0^{-1} D)^{-1}. \quad (11)$$

Interpretation: These covariance formulas are used to studentize the GMM estimator. The bootstrap has to reproduce this studentization to higher order.

2.6 The GMM t statistic

For testing the scalar null hypothesis

$$H_0 : \theta_r = \theta_{0r}, \quad (12)$$

let $\hat{\Sigma}_{rr}$ be the (r, r) component of the consistent covariance estimator. The t statistic is

$$T_{Nr} = \frac{N^{1/2}(\hat{\theta}_{Nr} - \theta_{0r})}{\hat{\Sigma}_{rr}^{1/2}}. \quad (13)$$

Under first-order asymptotics,

$$T_{Nr} \Rightarrow N(0, 1). \quad (14)$$

Interpretation: The paper studies symmetric two-sided tests based on $|T_{Nr}|$.

2.7 The overidentifying-restrictions statistic

Define

$$K_N(\theta) = \hat{\Omega}_N(\theta)^{-1/2} N^{1/2} \sum_{i=1}^N g(X_i, \theta). \quad (15)$$

The J statistic is

$$J_N = K_N(\hat{\theta}_N)' K_N(\hat{\theta}_N). \quad (16)$$

If $L_g > L_\theta$ and the overidentifying restrictions hold,

$$J_N \Rightarrow \chi_{L_g - L_\theta}^2. \quad (17)$$

Interpretation: The J test is often badly sized in finite samples. A main goal of the paper is to improve its critical values.

3 Why the bootstrap gives asymptotic refinements

3.1 First-order versus higher-order approximation

First-order asymptotic theory uses only the leading term of the limiting distribution. For example,

$$T_{Nr} \approx N(0, 1), \quad J_N \approx \chi^2_{L_g - L_\theta}. \quad (18)$$

Higher-order theory expands the distribution in powers of $n^{-1/2}$:

$$\mathbb{P}(T_{Nr} \leq z) = \Phi(z) + n^{-1/2}p_1(z)\phi(z) + n^{-1}p_2(z)\phi(z) + o(n^{-1}). \quad (19)$$

Interpretation: First-order critical values ignore the $n^{-1/2}$ and n^{-1} correction terms. If these are large, nominal 5% or 10% tests can have very different true levels.

3.2 Bootstrap logic

The bootstrap estimates the distribution of the statistic by resampling from the empirical distribution of the data. If the bootstrap distribution reproduces the Edgeworth expansion terms through order n^{-1} , then bootstrap critical values have smaller level errors than first-order asymptotic critical values.

Interpretation: The key is not that the bootstrap is exact. The key is that it estimates the higher-order terms that first-order theory ignores.

3.3 Why dependent data are harder

With independent observations, the bootstrap can resample observations. With dependent data, independent resampling destroys the time-series dependence structure.

Therefore the paper uses a block bootstrap:

- divide the sample into b nonoverlapping blocks;
- each block has length ℓ ;
- $b\ell = N$;
- resample b blocks with replacement.

Interpretation: Block resampling preserves local dependence inside each block, while using independent resampling across blocks.

3.4 Why naive bootstrap GMM statistics are wrong

Let X_i^* be the bootstrap sample. The naive bootstrap would compute the same statistic formulas using X_i^* and $\hat{\theta}_N^*$. Hall and Horowitz show this is not enough.

The reason is that the covariance of the bootstrap sample mean conditional on the data is not the same as the covariance estimator used in the original GMM statistic. Under block resampling, the conditional variance contains all within-block cross products, not only the lag terms that appear in the GMM covariance estimator.

Interpretation: If the bootstrap statistic is not corrected, its conditional distribution does not match the sample statistic distribution to the order needed for asymptotic refinements.

4 Bootstrap construction in detail

4.1 Recentered bootstrap moments

The bootstrap version of the moment condition is recentered. Define

$$g^*(X_i^*, \theta) = g(X_i^*, \theta) - E^*[g(X_i^*, \hat{\theta}_N)], \quad (20)$$

where E^* denotes expectation under the bootstrap distribution conditional on the original sample.

Then the bootstrap population moment condition is

$$E^*[g^*(X_i^*, \theta)] = 0. \quad (21)$$

Interpretation: Recentering is essential in overidentified models. Without it, the empirical bootstrap population generally does not satisfy the model moment restriction at any parameter value.

4.2 Bootstrap GMM estimator

The fixed-weight bootstrap estimator solves

$$\hat{\theta}_N^* = \arg \min_{\theta \in \Theta} \left[N^{-1} \sum_{i=1}^N g^*(X_i^*, \theta) \right]' \Omega \left[N^{-1} \sum_{i=1}^N g^*(X_i^*, \theta) \right]. \quad (22)$$

The two-step bootstrap estimator solves an analogous criterion using the bootstrap covariance matrix computed from g^* .

Interpretation: The bootstrap estimator is constructed so the resampled problem mimics the original GMM problem, but centered around the sample estimate rather than the unknown true parameter.

4.3 Corrected bootstrap t statistic

Let $\hat{\Sigma}_N^*$ be the bootstrap analog of the usual covariance estimator. Let $\hat{\Sigma}_N^b$ denote the covariance of the bootstrap asymptotic distribution conditional on the data. The corrected bootstrap t statistic rescales the naive statistic so its conditional variance is one:

$$T_{Nr}^* = \frac{N^{1/2}(\hat{\theta}_{Nr}^* - \hat{\theta}_{Nr})}{(\hat{\Sigma}_N^b)^{1/2}_{rr}}. \quad (23)$$

Interpretation: In the paper's notation, the correction is expressed through matrices W_N , $\hat{\Omega}_N$, D_N , and $\hat{\Sigma}_N^b$. The economic idea is simple: studentize by the covariance generated by the block bootstrap, not by the covariance formula that applies to the original statistic.

4.4 Corrected bootstrap J statistic

The corrected bootstrap version of $K_N(\hat{\theta}_N)$ premultiplies the bootstrap sample moment vector by a matrix that makes its conditional covariance match the covariance required for the J statistic. In the paper's notation,

$$K_N^*(\hat{\theta}_N^*) = (V_N^+)^{1/2} \hat{\Omega}_N (\hat{\theta}_N^*)^{-1/2} N^{1/2} \sum_{i=1}^N g^*(X_i^*, \hat{\theta}_N^*), \quad (24)$$

where V_N^+ is a Moore-Penrose generalized inverse of the covariance correction matrix.

The bootstrap J statistic is

$$J_N^* = K_N^*(\hat{\theta}_N^*)' K_N^*(\hat{\theta}_N^*). \quad (25)$$

Interpretation: The covariance correction is algebraically more complicated than for the t statistic because the J statistic is a quadratic form in moment restrictions after projection away from estimated parameters.

4.5 Bootstrap critical values

For a nominal level α :

- the t test uses the $(1 - \alpha)$ quantile of the bootstrap distribution of $|T_{Nr}^*|$;
- the J test uses the $(1 - \alpha)$ quantile of the bootstrap distribution of J_N^* .

Interpretation: These critical values replace the normal and chi-square critical values used in first-order asymptotic inference.

5 Regularity conditions

5.1 Assumption 1: dependence through exponentially decaying innovations

The process X_i is represented as a measurable function of iid innovations:

$$X_i = h(\varepsilon_i, \varepsilon_{i-1}, \varepsilon_{i-2}, \dots). \quad (26)$$

The effect of remote innovations decays exponentially.

Interpretation: This implies strong mixing at an exponential rate. It permits dependent data while ensuring that distant observations are weakly related.

5.2 Assumption 2: identification and compactness

The true parameter θ_0 is an interior point of a compact parameter set Θ and is the unique solution to

$$E[g(X, \theta)] = 0. \quad (27)$$

Interpretation: This rules out weak identification and boundary complications.

5.3 Assumption 3: moments, smoothness, and tails

The moment function $g(X, \theta)$ has finite expectation, is Lipschitz in θ , and is four-times differentiable near θ_0 . Its derivatives and Lipschitz constants have sufficiently thin tails.

Interpretation: These conditions allow Taylor expansions of the estimator and statistic through high enough order to prove $o(n^{-1})$ refinements.

5.4 Assumption 4: smoothness of the expanded moment vector

The vector $f(X_i, \theta)$ collects components of moments, covariance terms, and derivatives through order four. The assumption requires Lipschitz continuity and tail bounds for this expanded vector.

Interpretation: The proof works by writing statistics as smooth functions of a finite set of sample moments. Assumption 4 controls those moments.

5.5 Assumption 5: continuous component

The random vector X has at least one continuously distributed component, and certain derived variables are absolutely continuous conditional on the discrete components.

Interpretation: A continuity condition is needed for Edgeworth expansions and Cramer-type arguments.

5.6 Assumption 6: dependent-data Cramer condition

The paper imposes a strengthened Cramer condition for dependent data. It restricts the characteristic function of sums of the expanded moment vector over blocks.

Interpretation: This is one of the strongest and least transparent assumptions. Its role is technical: it ensures the existence of Edgeworth expansions with dependent data.

5.7 Block length condition

For the main bootstrap theorem, block length must satisfy

$$\ell = N^\gamma, \quad \frac{11}{50} < \gamma < \frac{12}{50}. \quad (28)$$

Interpretation: The block length must increase with sample size, but not too quickly. It must be long enough to capture dependence and short enough to keep enough approximately independent blocks.

6 Main theoretical results

6.1 Theorem 1: Edgeworth expansions for the sample statistics

Theorem 1 states that, under Assumptions 1–6, the sample t statistic T_{Nr} and the sample J statistic J_N admit Edgeworth expansions through order n^{-1} .

For the t statistic, the distribution can be written schematically as

$$\mathbb{P}(T_{Nr} \leq z) = \left[1 + N^{-1/2}\pi_1(D, v) + N^{-1}\pi_2(D, v) \right] \Phi(z) + o(n^{-1}), \quad (29)$$

where $D = d/dz$ and the π_j are differential-operator polynomials whose coefficients depend on moments of the data.

For the J statistic,

$$\mathbb{P}(J_N \leq z) = \int_{-\infty}^z \left[1 + N^{-1}\pi(D, v) \right] dF_{\chi^2_{L_g - L_\theta}}(x) + o(n^{-1}). \quad (30)$$

Interpretation: The theorem formalizes the higher-order distributional error in first-order normal and chi-square approximations.

6.2 Theorem 2: Edgeworth expansions for bootstrap statistics

Theorem 2 states that the corrected bootstrap statistics T_{Nr}^* and J_N^* have conditional Edgeworth expansions through order n^{-1} , except on a set whose probability is $o(N^{-1})$.

Interpretation: The bootstrap reproduces the same type of expansion as the sample statistic. This is the key technical step behind bootstrap refinements.

6.3 Theorem 3: bootstrap critical values have refined levels

Let $z_{T\alpha}^*$ and $z_{J\alpha}^*$ be the bootstrap critical values for the symmetric t test and the J test. Under the null hypothesis,

$$\mathbb{P}(|T_{Nr}| > z_{T\alpha}^*) = \alpha + o(n^{-1}), \quad (31)$$

and, if the overidentifying restrictions hold,

$$\mathbb{P}(J_N > z_{J\alpha}^*) = \alpha + o(n^{-1}). \quad (32)$$

Interpretation: This is the central result. Bootstrap critical values reduce size error relative to first-order asymptotic critical values.

6.4 What the theorem does not say

- It does not say the bootstrap test is exact in finite samples.
- It does not say the bootstrap always removes all level distortions.
- It relies on strong smoothness, tail, and dependence assumptions.
- It requires corrected bootstrap statistics, not naive bootstrap GMM statistics.

Interpretation: The theorem is a higher-order asymptotic result, not a finite-sample guarantee.

7 Monte Carlo design and evidence

7.1 Simulation procedure

Each Monte Carlo experiment proceeds as follows:

1. Simulate data from a specified nonlinear moment-condition model.
2. Compute the two-step GMM estimator, the J statistic, and the symmetric t statistic.
3. Determine rejections using first-order asymptotic critical values.
4. Compute bootstrap critical values using 100 bootstrap replications.
5. Determine rejections using bootstrap critical values.
6. Repeat the experiment 1,000 times to estimate empirical test levels.

Interpretation: The simulations measure test size, not power. They ask whether nominal 10%, 5%, and 1% tests reject at the correct frequency under the null.

7.2 Nonlinear moment-condition model

The model is a simplified asset-pricing moment system:

$$E \{ \exp[\mu - \theta(X + Z) + 3Z] - 1 \} = 0, \quad (33)$$

$$E \{ Z (\exp[\mu - \theta(X + Z) + 3Z] - 1) \} = 0. \quad (34)$$

The true value is

$$\theta_0 = 3. \quad (35)$$

The variable X is sampled independently from $N(0, s^2)$, where $s = 0.2$ or $s = 0.4$. The variable Z is independent of X and has marginal distribution $N(0, s^2)$, but it is either independent across time or follows an AR(1) process with first-order serial correlation $\rho_z = 0.75$.

Interpretation: The model is nonlinear and overidentified, so it is a useful laboratory for t and J tests based on GMM.

7.3 Sample sizes and block lengths

The experiments use

$$n \in \{50, 100\}. \quad (36)$$

When $\rho_z = 0$, the number of blocks is n , which is equivalent to individual resampling. When $\rho_z > 0$, the number of blocks is $n/5$ or $n/10$.

Interpretation: The design compares independent and dependent cases and checks whether block length matters in finite samples.

7.4 Main Monte Carlo finding

The empirical levels of tests using asymptotic critical values are often much larger than nominal. Bootstrap critical values usually reduce these size distortions, especially for the symmetric t test.

- At nominal 10% and 5% levels, bootstrap critical values often reduce t -test level distortions by 40–90%.
- Bootstrap corrections are useful but do not fully eliminate size errors.

- Size distortions are smaller when $s = 0.2$ than when $s = 0.4$.
- Block length has little effect on empirical levels in the reported experiments.

Interpretation: The simulation confirms the theoretical message: the bootstrap improves critical values, but it is not a magic finite-sample fix.

8 Tables

8.1 Table I: Levels of t and J tests

Read:

- The nominal levels are 10%, 5%, and 1%.
- For $n = 50$, $s = 0.4$, and independent data, the asymptotic t test rejects at 26.0%, 19.3%, and 11.4%, far above nominal levels.
- In the same row, the bootstrap t test rejects at 16.5%, 12.3%, and 6.2%, still oversized but much closer to nominal.
- For the J test, asymptotic critical values also produce oversizing, and bootstrap critical values reduce some but not all distortions.
- The improvement is stronger for the t test than for the J test.
- Dependent data cases with $\rho_z = 0.75$ show that block bootstrap critical values can improve inference under serial dependence.

Economic message: The table gives the paper’s practical message. First-order critical values can be badly oversized in GMM, and bootstrap critical values usually improve finite-sample size.

TABLE I
LEVELS OF T AND J TESTS FOR A NONLINEAR MODEL WITH ASYMPTOTIC AND BOOTSTRAP
CRITICAL VALUES^a

n	s	r_z	b	t Test with Asymp. Crit. Values			t Test with Boot. Crit. Values			J Test with Asymp. Crit. Values			J Test with Boot. Crit. Values		
				0.10	0.05	0.01	0.10	0.05	0.01	0.10	0.05	0.01	0.10	0.05	0.01
50	0.2	0.0	50	0.153	0.092	0.042	0.104	0.063	0.031	0.124	0.088	0.022	0.123	0.082	0.032
50	0.2	0.75	5	0.194	0.144	0.073	0.150	0.104	0.064	0.143	0.093	0.052	0.138	0.096	0.058
		0.75	10				0.147	0.092	0.057				0.126	0.078	0.046
50	0.4	0.0	50	0.260	0.193	0.114	0.165	0.123	0.062	0.228	0.171	0.102	0.197	0.149	0.095
50	0.4	0.75	5	0.242	0.180	0.099	0.156	0.125	0.069	0.217	0.168	0.113	0.183	0.140	0.109
		0.75	10				0.182	0.131	0.086				0.198	0.164	0.122
100	0.2	0.0	100	0.155	0.090	0.024	0.122	0.067	0.023	0.119	0.070	0.026	0.116	0.068	0.027
100	0.2	0.75	10	0.182	0.110	0.050	0.133	0.080	0.036	0.139	0.090	0.032	0.138	0.093	0.045
		0.75	20				0.125	0.065	0.026				0.124	0.089	0.045
100	0.4	0.0	100	0.224	0.150	0.073	0.161	0.102	0.054	0.194	0.132	0.067	0.164	0.113	0.063
100	0.4	0.75	10	0.216	0.150	0.086	0.141	0.096	0.057	0.188	0.132	0.072	0.156	0.124	0.076
		0.75	20				0.149	0.110	0.064				0.172	0.126	0.083

^a Underline indicates that the empirical level is not significantly different from the nominal level at the 0.01 level.

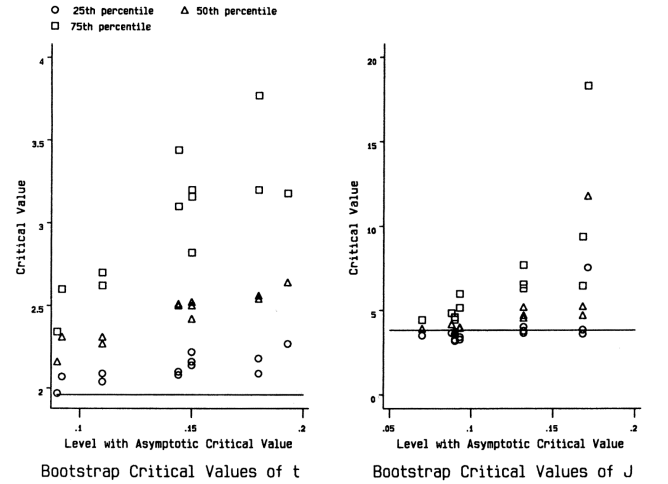


FIGURE 1—Percentiles of distributions of bootstrap critical values of t and J tests versus empirical levels based on asymptotic critical values.

8.2 Figure 1: Bootstrap critical values as diagnostics

Read:

- The figure plots percentiles of bootstrap critical values against empirical levels produced by asymptotic critical values.
- For the t test, bootstrap critical values rise when the asymptotic test is more oversized.
- This means the bootstrap can diagnose when first-order asymptotic critical values are too small.
- For the J test, the diagnostic relation is weaker but still informative at the median percentile.

Economic message: Bootstrap critical values do more than replace asymptotic critical values. Their distance from first-order critical values can signal that first-order asymptotics are unreliable.

9 Appendix and proof structure

9.1 Step 1: Approximate statistics by smooth functions of sample moments

The proofs first show that the GMM estimators and statistics can be approximated by smooth functions of sample moments with error $o_p(n^{-1})$.

Interpretation: This allows the authors to apply Edgeworth expansion results for smooth functions of sample averages.

9.2 Step 2: Establish Edgeworth expansions

The paper then applies results from Bhattacharya, Chandra and Ghosh, and Gotze and Hipp to obtain Edgeworth expansions for dependent data.

Interpretation: This is where the strong mixing, tail, differentiability, and Cramer assumptions are used.

9.3 Step 3: Show bootstrap expansions match sample expansions

The bootstrap statistics are shown to have conditional expansions with the same polynomial terms as the sample statistics, up to the required order.

Interpretation: Matching expansion terms is what yields the $o(n^{-1})$ level error for bootstrap critical values.

9.4 Role of the propositions

The appendix propositions establish stochastic expansions for the original and bootstrap GMM statistics. They show that the statistics can be expressed as smooth functions of centered sample or bootstrap moments plus negligible remainder terms.

Interpretation: The propositions are the technical bridge between the GMM optimization problem and the Edgeworth-expansion machinery.

10 Short summary

- GMM tests can have large finite-sample size distortions when first-order asymptotic critical values are used.
- The paper studies the GMM t test and the overidentifying-restrictions J test.
- The key theoretical tool is an Edgeworth expansion through order n^{-1} .
- With dependent data, the bootstrap must resample blocks rather than individual observations.
- Bootstrap moments must be recentered because the empirical bootstrap population generally does not satisfy overidentified moment restrictions exactly.
- Naive bootstrap versions of the t and J statistics do not have the right higher-order distributions under block resampling.

- Hall and Horowitz derive corrected bootstrap statistics that account for the covariance induced by block resampling.
- Under strong regularity conditions and suitable block length, bootstrap critical values produce tests with level $\alpha + o(n^{-1})$.
- Monte Carlo simulations show that bootstrap critical values usually reduce size distortions, especially for symmetric t tests.
- Bootstrap critical values can also act as diagnostics for the reliability of first-order asymptotic critical values.

11 Conclusion

11.1 Core conclusion

Hall and Horowitz provide a higher-order theory for bootstrap critical values in GMM testing. They show that the bootstrap can improve the accuracy of critical values for both t tests and the J test, including in dependent-data settings, but only if the bootstrap statistics are constructed correctly.

11.2 Economic and econometric intuition

The econometric intuition is that first-order normal and chi-square approximations ignore higher-order skewness, kurtosis, and studentization effects. The bootstrap estimates these higher-order terms from the sample. With dependent data, however, block resampling changes the covariance structure of bootstrap sample moments, so the statistic must be adjusted to mimic the distribution of the original statistic.

11.3 Takeaway

The paper is a foundational result for bootstrap GMM inference. It explains both why bootstrap critical values can be better than asymptotic ones and why block-bootstrap GMM requires special care. The practical lesson is clear: when using GMM with finite samples, especially under dependence, bootstrap inference can be valuable, but the bootstrap statistic must be theoretically aligned with the statistic being approximated.